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School of Engineering

A Bird of Prey Inspired Robotic Gripper with Adjustable Digit Separation

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Abstract

This project presents a bird-inspired robotic gripper designed to study how the angle of digit-separation influences object deformation during grasping. A rigid, tendon-driven 3D-printed claw was built, actuated by nylon tendons and compact DC gear motors, with a vernier mechanism enabling precise adjustment of digit angles. Tests on PLA and TPU substrates found no consistent relationship between digit-separation angle and substrate deformation. Analysis indicated that high tendon friction and insufficient motor torque limited performance, while inconsistencies in the design limited reliability. Despite this, the prototype validated a functional biomimetic concept and establishes groundwork for future haptic grippers with improved actuation and sensing systems.

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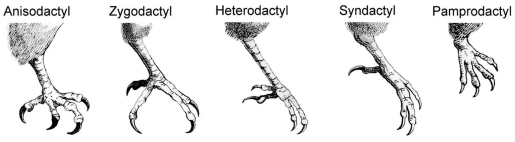
1. Introduction

Grasping objects without causing unwanted deformation is a persistent challenge in robotics [1]. We hypothesise that a bird-inspired robotic gripper has an optimal digit-separation angle that minimises deformation, avoiding the need to simply raise grip force.

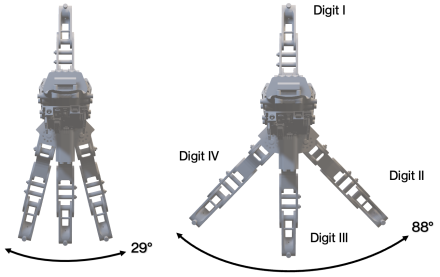
This work explores the use of bird-inspired geometry in a rigid, talonless, tendon-driven gripper. Soft robotics are widely used to handle damage-prone items because flexible surfaces conform to the object's shape and distribute pressure [2]. Our design aims to address this challenge of limiting deformation, but does so through bird-inspired geometry rather than material softness. Our gripper uses a tendon-driven mechanism to mimic how birds achieve secure, low damage grasps through coordinated tendon motion and digit arrangement [3]. We focus on the digit-separation angle θ_d that can be seen in Figure 1b.

This study tests how the deformation of a cylinder varies when gripped at different values of θ_d. The gripper closes on the same cylindrical object each time using the same actuation settings.

Biomimetics, from the Greek words 'bios' (life) and 'mimesis' (imitation), is the study of nature and natural phenomena to understand the principles of underlying mechanisms, to obtain ideas from nature, and to apply concepts that may benefit science, engineering and medicine [4]. Building on this method, we drew from birds' toe arrangements when designing our claw. In nature, bird claws have converged to five prevalent arrangements:



(a) Prevalent Bird Claw Arrangements [5]



(b) Visualisation of Digit Separation Angle, θ_d

Figure 1: Overview of our bird claw parameters

Our research outlined that the anisodactyl (aniso = unequal, dactyl = toe) arrangement exhibited the greatest grip strength for its size [6]. Pairing this fact with the simplified design associated with anisodactyl arrangements, where digits I and III can be modelled the same, and digits II and IV the same (Figure 1a) [7], it was selected as the most suitable configuration. With time being a significant restriction in this project, simplicity in design and geometry which lends itself to high grip strength were desired qualities.

Further research explored other uses for bird-inspired robotic claws, such as landing and perching [8] and grasping objects of various shapes and weights [9]. This highlighted a gap in current knowledge: there is **no study that systematically varies the angle of digit separation in a bird-inspired robotic claw to measure substrate deformation**.

2. Methods

To gather data to inform our hypothesis' validity, we built a small prototype claw with 3D-printed digits. The gripper uses fishing lines as tendons and four DC gear motors. There is a one-to-one mapping: one tendon and one motor for each digit. The digit separation angle is set with a vernier adjuster, and a force-sensing resistor is mounted onto the palm to provide haptic feedback.

The concept focuses on exploring whether adjusting the separation of digits alone can achieve secure, deformation-free grasping inspired by biology. The adjustable-digit framework provides potential groundwork for future development and implementation into industries that require gentle and precise handling.

2.1. Design

2.1.1. Mechanical Design

The overall mechanical design of the claw (Figure 2) achieved a compact layout that accommodated all components within a small footprint while maintaining access for assembly and maintenance. The claw was not faultless, but allowed us to adjust the angle of digit separation, and to carry out grasping tests on cylindrical substrates.

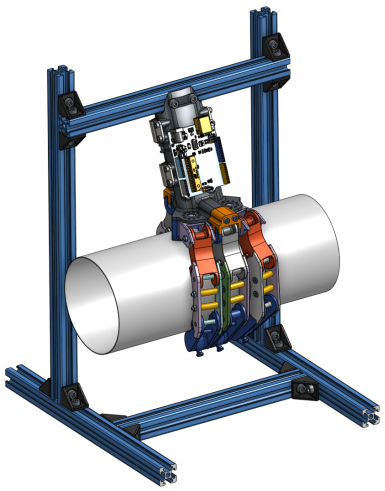
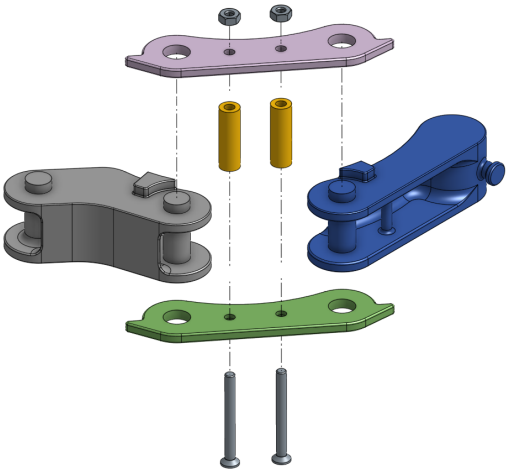
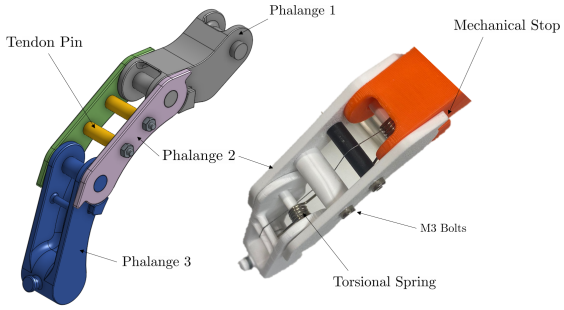


Figure 2: Overall Claw Assembly

Each digit was designed so that, in the fully closed position, the claw formed an enveloping grasp around a 100 mm diameter cylinder (see the grasp performed in Figure 2). To achieve this, each digit is made from 3 phalanges as seen in Figure 3b. To perform a successful actuation of a tendon-driven gripper, two motions are required: flexion and extension, i.e., closing and opening, respectively.



(a) Digit Assembly Exploded View



(b) CAD Digit and Manufactured Digit

Figure 3: Digit Design

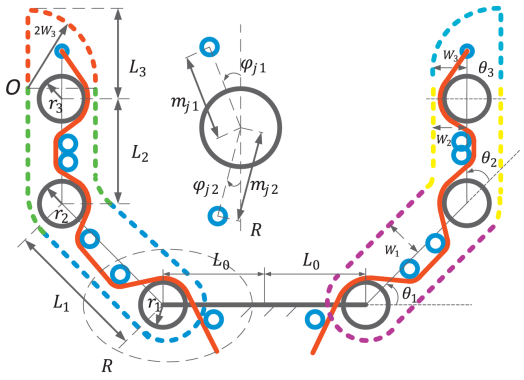
The flexion motion is achieved by routing a tendon around joints and tendon pins (Figure 4) such that when the tendon experiences tension, the digits curl and the claw grips. The extension mechanism can be achieved either by use of joint (torsional) springs, or by use of a second tendon kept under constant tension using linear extension springs (See The Velo Gripper [10] and Figure 4 for details). We originally opted to use an extensor tendon as using torsional springs constrains r_1, r_2 and r_3 (Figure 4) which gives less flexibility in the design of the tendon routing mechanism.

After a first trial (see design iteration A in the supplementary material) we opted to change the design to utilise torsional springs. The justification for this is that the extensor tendon requires pre-tension; this, in turn, requires the linear extension springs to be extended even when the claw is in its fully open position. Given the large range of motion of the digits, we were unable to find springs that were stiff enough, and importantly, short enough to maintain this tension for all claw positions between open and closed.

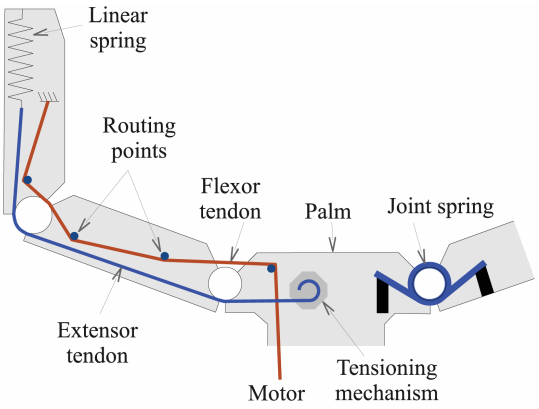
Between adjacent phalanges there is a mechanical stop to prevent the digit from overextending during extension.

There is extensive multivariable optimisation that can be performed [11] to determine the ideal placement, and radius of each 'pulley' in this system, however, given our time constraints, we took inspiration from existing research [10], [11] as our baseline, and used a trial-and-error approach to achieve a mechanism that flexed and extended in the way we desired. One key constraint during this process was that an M3 bolt passes through the centre of each tendon pin, holding the 3 phalanges together. These bolts also hold the tendon pins in place. As such the minimum radius of a tendon pin must be the sum of an M3 clearance hole radius, plus an acceptable wall thickness, which we determined through experimentation as 0.8mm, meaning 2 wall layers extruded from a 0.4mm nozzle.

The minimum radius is therefore $r_{\min} = \frac{3.2}{2} + 0.8 = 2.4$ mm.



(a) Tendon routing schematic [10]



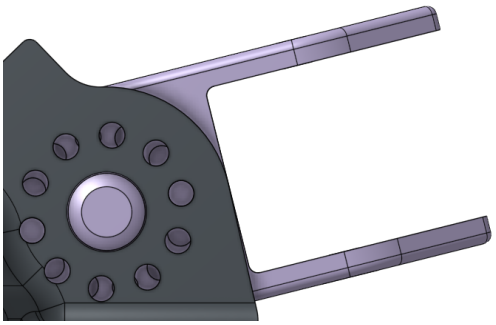
(b) Tendon routing design with motor and springs [11]

Figure 4: Tendon Routing Configurations from Related Studies

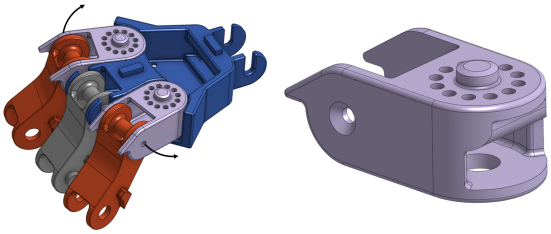
The design of the palm, knuckles, and leg (Figures 5, 6a and 6b) should be discussed in parallel, since they are closely interlinked. The knuckles are secured in place between the palm and the leg, which are bolted together with 2 M3x30mm countersunk bolts. Two rectangular pins protrude from the top of the palm that act as alignment features, fitting into complementary slots on the underside of the leg.

Each knuckle can rotate 30° from an angle of 14° to 44° thus allowing θ_d to be varied from 28° to 88°. The knuckles are physically unable to rotate outside of this range, as the palm is designed such that the knuckle's outer surface is flush with a physical stop in its most adducted and most abducted positions.

To secure the knuckle at a specific angle, a vernier mechanism is used. On the top face of the knuckle, there are 12 equally spaced M2.5 through holes concentric about the knuckle's rotation axis. On the leg there are a further 10 equally spaced M2.5 through holes also concentric about the knuckle's rotation axis. As such, every 6° of rotation, one pair of these holes align (see Figure 5a). An M2.5 bolt can be placed through the aligned holes, fixing θ_d.



(a) Vernier Mechanism



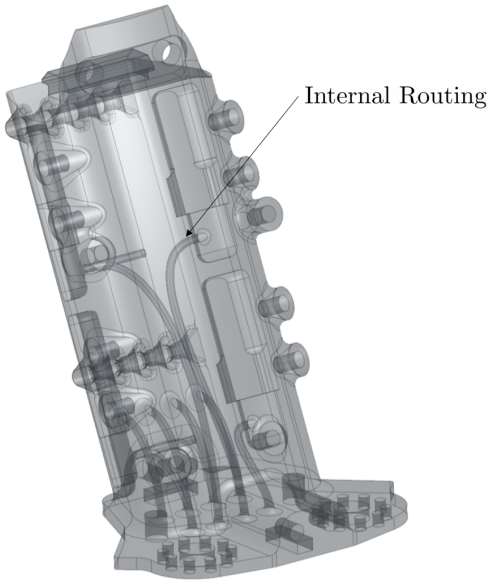
(b) Palm and Knuckle

Figure 5: Palm with Digit Angle Adjustment

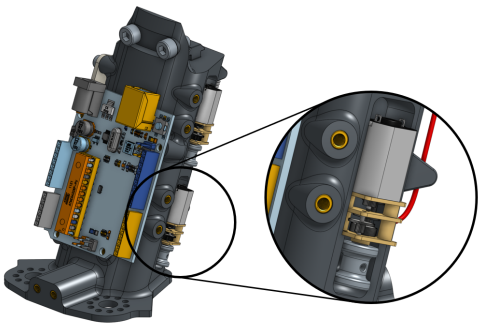
To actuate the digits, we used braided fishing line as our tendons, since it has high tensile strength, and low stretch resulting in low latency actuation without breaking. Whilst the line performs well in tension in its undamaged state, as soon as the line gets snagged or frayed, it fails almost immediately under tension. As such, we made an effort to ensure that the path the tendon takes between the digit tip and the motor is smooth with no sharp corners or edges. This can be seen at the exit of the knuckle (Figure 5) where there is a large radius around which the tendon turns before it enters the leg.

3D printing of the leg structure allowed us to route the tendons through the centre of the leg, thereby taking a short path (which lowers overall stretch) and keeps each tendon separate, and in a controlled channel to avoid tangling. The internal routing channels can be seen in Figure 6a. These were modelled in OnShape as splines for which the start and end of each spline is normal to the hole through which the tendon enters/exits. This is once again a precaution to prevent the wire from snagging or fraying.

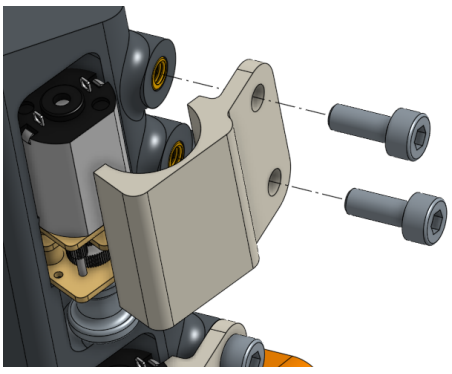
A downside of this routing method is that it introduces significant amount of friction between the fishing line and the channels, since they are in constant contact. Given that our motors are low torque, this friction subtracts a significant proportion of gripping force from each digit. Had we more time, we could have characterised this frictional loss by hanging a mass from the end of a tendon, and measured its vertical acceleration as the motor is actuated. This would've allowed us to compare the actual force to the expected force, thus deriving how much force is being lost to friction.



(a) Transparent-Leg Showing Internal Fishing Line Routing



(b) Motor Embedded Within Leg Structure



(c) Motor Covers

Figure 6: Leg Design

Despite the frictional losses, the internal tendon routing was still a success overall from a design standpoint, as the convenience of having all the tendons internal to the structure is vital. This first prototype has high friction in the channels because we 3D printed it from PLA with a 0.4mm layer height.

The leg structure also houses the motors that actuate the claw. Figure 6c shows how each motor is held in place by a bolted cover. We made use of heat-set M3 inserts to allow parts to be bolted to the leg, namely the motor covers, the palm, the Arduino and the covers for digits I and III. Attached to the D-shaft of each motor is a spool we printed from TPU with a shore hardness of 68D. Since TPU is an elastomer, we predicted that the tendon would grip TPU better than PLA since its elastomeric surface would allow for more mechanical interlocking. We confirmed this from testing both varieties of spools, and used the TPU spools for our final prototype.

2.1.2. Electrical and Software Design

The electrical subsystem, detailed in Table 1, provided actuation and sensing capabilities for the gripper. Component selection prioritized cost-effectiveness, compatibility, and ease of integration within the project's £150 budget and short timeline.

Component	**Key Specifications**
Arduino Uno Rev3	ATmega328P microcontroller 10-bit ADC (0–1023) 5V reference voltage 4.88 mV resolution
DRI0009 Controller	L298N dual H-bridge 4-channel independent control 2A max per channel PWM speed control
N20 Micro Motor	Operating voltage: 6V Gear ratio: 298:1 Stall torque: 3.3 kg·cm Stall current: 700 mA No-load speed: 60 RPM
Interlink 402 FSR	Active area: 12.7 mm diameter Force range: 0.1–10 N Resistance (unloaded): >10 MΩ Resistance (loaded): 200–1000 Ω

Table 1: Electrical System Component Specifications

N20 micro gear motors were selected for finger actuation due to their compact form factor and favourable torque-to-size ratio given the gear reduction. The Arduino Uno Rev3 was chosen as the microcontroller platform for its widespread adoption, extensive documentation, and native compatibility with the DFRobot motor shield. The motor shield itself, based on the L298N dual H-bridge integrated circuit, enabled independent control of all four motors through a simplified interface that abstracted the underlying H-bridge switching logic [12]. Each motor was controlled via two pins: a digital pin for direction (forward/backward) and a PWM-capable pin for speed modulation.

A single force-sensitive resistor (FSR) was mounted on the palm to provide rudimentary force feedback. The FSR's voltage divider configuration (Equation 1) converted resistance changes to measurable voltages:

$$V_{out} = V_{cc} \cdot \frac{R_{FSR}}{R_{FSR} + R_{fixed}} \tag{1}$$

where $V_{cc} = 5$ V and $R_{fixed} = 10$ kΩ. This resistor value was selected to match the logarithmic midpoint of the sensor's resistance range (200 Ω to >10 MΩ), ensuring maximum voltage swing across the ADC's input range, and to satisfy the ATmega328P's maximum 10 kΩ source impedance requirement for accurate analog-to-digital conversion [13]. However, the FSR ultimately proved unreliable during testing due to inherent drift and sensitivity limitations, and was not used in final experiments.

The control architecture employed open-loop PWM for motor speed control, with no position feedback due to the absence of encoders. The Arduino firmware implemented a serial command protocol operating at 9600 baud, accepting ASCII commands from a Python-based host interface. Commands followed the format `M<motor><direction><speed><duration>`, enabling timed or continuous operation of individual motors or all motors simultaneously. For example, `M1F150C` drove motor 1 forward at speed 150 continuously, while `M0B2003000` drove all motors backward at speed 200 for 3000 ms. Emergency stop commands (`S<motor>`) immediately shutdown the motors. The Python interface provided an interactive command-line menu for operator control, translating user inputs into properly formatted serial commands. Firmware and control software are provided in the supplementary materials.

2.1.3. Material Selection

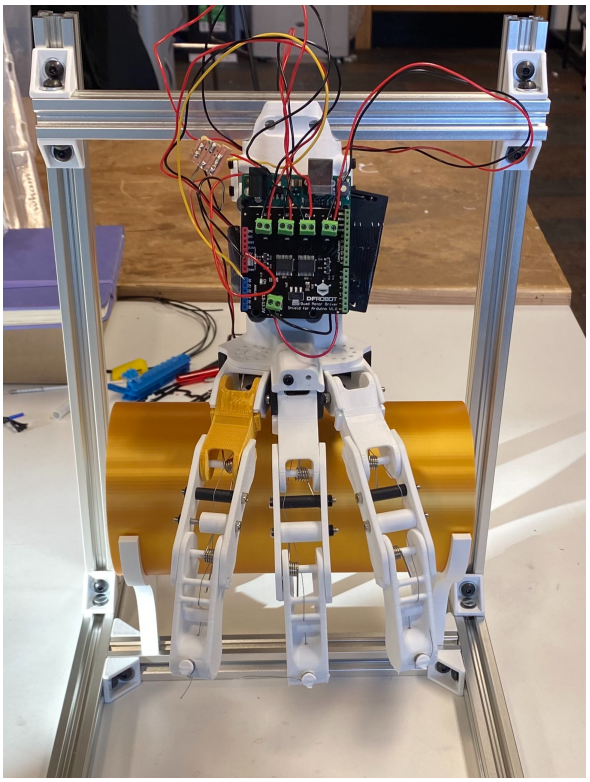
For the tendon, a 4-strand braided ultra-high-molecular-weight polyethylene (UHMWPE) fishing line runs from the digit tips through the finger segments and palm to the leg-mounted motor. A 10-lb line (rated break ≈ 44.5 N) was chosen for its small diameter and low stretch, [14]. We opted for the 4-strand because its slightly rougher texture holds better onto surfaces and resists scuffing [15]. We pre-tensioned each tendon before fitting it. We fasten the tendons to the spools and digit tips using simple knots [16]. These are the same good habits used with Dyneema/HMPE ropes, and they apply here because PE braid is made from the same UHMWPE fibre[14].

The overall assembly was constructed using the materials listed in Table 2.

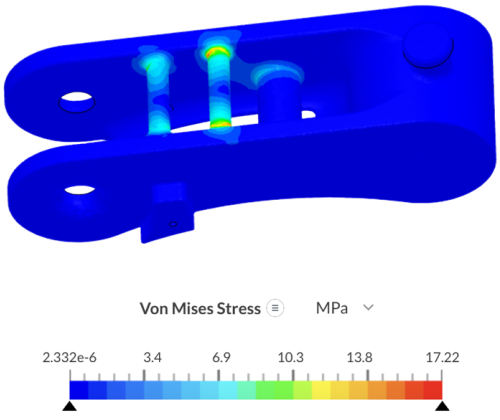
Resource	Quantity	Datasheet
PLA Filament	450 g	Link
TPU Filament	75 g	Link
PETG Filament	50 g	Link
M3 Heat Set Inserts	16 pcs	Link
Aluminium Extrusion	1760 mm	Link
Torsional Springs	12 pcs	Link
M3×30 mm Bolts	18 pcs	Link
Braided Fishing Line	120 mm	Link

Table 2: Mechanical resources used in assembly.

PLA was chosen as the primary material for all claw components. The exception being the palm, and the digit covers which are PETG, not for particular material considerations, but because we ran out of PLA. PLA is a good choice for our application as it prints reliably and quickly on the Bambulab A1 and X1C which were the printers we had available to us during this project.



(a) Final Prototype



(b) Structural FEA to Validate Material Choice

Figure 7: Assembly and Material Analysis

To support our material choice, a static structural FEA study was carried out in Simscale, on the 3rd phalange, since this part was identified as having the thinnest tendon pins. Visual results of the study can be seen in Figure 7b. A mesh convergence study and details of the boundary conditions can be found in our supplementary materials folder.

2.2. Experimental Set Up and Implementation

The experiment required a modular test stand which was assembled using: 20-series aluminium extrusions, PLA brackets and M5 Hardware. Aluminium extrusions were used for the main frame since their T-slot design enables modularity, allowing brackets and fixtures to be easily repositioned. PLA was chosen for the brackets and holders owing to its low cost and ease of 3D printing. Although its Young's modulus (∼3.5 GPa) and density (1.25 g/cm^3) [17, 18] are much lower than aluminium, at 100% infill it provides sufficient rigidity for static loading. The PLA holders were designed to secure the substrate, ensuring repeatability between tests.

M5 hardware was used for compatibility with the extrusion slots and to simplify reconfiguration. A ruler mounted along the frame and a horizontally aligned camera provided a visual reference for accurate deformation measurements. This test stand with the claw assembled can be seen in Figure 7a.

2.3. Tests

The testing procedure was as follows:

1. The substrate was placed in the test stand.
2. The digits were normalised such that the test started with all digits at the same height, and all tendons under tension. This was checked with a spirit level.
3. A camera was set up to film through the bore of the cylinder, behind which a ruler was placed vertically.
4. The claw was actuated, and would grip the cylinder until the stall torque of the motor was reached.
5. The final diameter of the cylinder was measured using the footage from the camera.
6. The motors unwind, and the claw extends to its original position.
7. The deformation of the cylinder was calculated as, deformation, $\Delta D = D_{\text{before}} - D_{\text{after}}$.

Testing substrates of different material ensured that the results and conclusions would not be material-specific.

These substrates were:

Black Cylinder - PLA, 0.4mm Wall Thickness
White Cylinder - TPU, 0.8mm Wall Thickness

3. Results

The data in Figure 8 is insufficient for us to determine the validity of our hypothesis. We would expect the black cylinder to deform more than the white cylinder at every value of θ_d, however from our collected data this only happens 5 of 6 times. If the data points in Figure 8 represented mean values of 10 or more tests, it would be fair to conclude that this trend is present, however since we only have a single datum at each value of θ_d for each cylinder, it is not possible to confidently say that the black cylinder consistently was deformed more. This test would have served as our sanity check to test that the claw is gripping consistently, so not having this validation is the first reason we failed to accept our hypothesis.

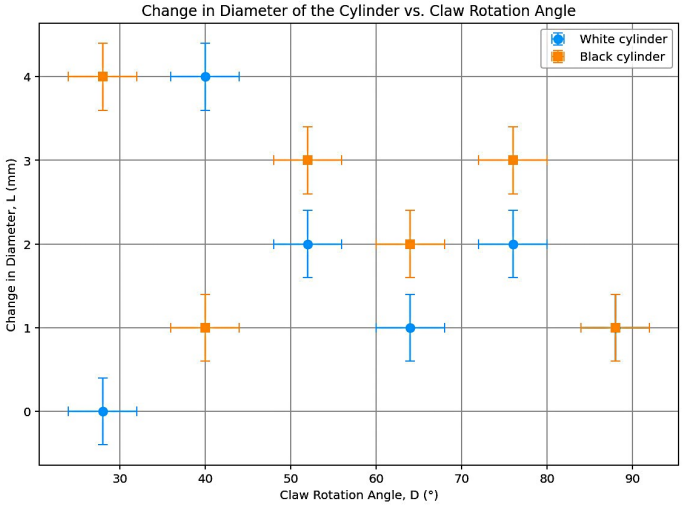


Figure 8: Change in diameter ΔD vs. digit-separation angle θ_d

Secondly, we analysed the data for a potential minimum, in order to prove our hypothesis correct. To say there **is** an optimal angle we would need to have evidence that at one particular θ_d, the deformation of the cylinder was minimised for both substrates. As is evident from Figure 8 no such or minimum exist.

Importantly, **had there been a clear trend in this data with a minimum point we still would have failed to accept our hypothesis**. Given our lack of data, and lack of confidence in the data that was collected, any trend visible in Figure 8, could be attributed entirely to coincidence.

Finally, it is worth commenting on the phrasing of 'There exists an **optimal** angle' from our hypothesis. While in the case of our research we say 'optimal' to mean minimum deformation, if we had suitable data to suggest that a particular angle, or range of angles provided more consistent deformation, i.e., deformed both substrates the same amount with little variance between tests, we may have been able to form an alternate conclusion: that there is an optimal angle for **controlling** indentation. However, as previously stated, the data is insubstantial to draw such a conclusion.

4. Discussion

4.1. Evaluation of Approach

Comparing our work to existing research that aimed to optimise robotic grippers [8][10][11], our design lacked in its ability to produce replicable tests, and substantial data. We believe our final prototype achieves Technical Readiness Level 4 (TRL4), but since it failed to collect the data we desired, we do not meet TRL 5 which mandates a validated design.

Given our small budget and fast timeline, it was ambitious to make the claw as small as we did. Having our design be more compact, meant we had to make compromises such as not utilising an extensor tendon, and the low budget meant we couldn't afford motors with higher torque.

Our approach of gripping a cylinder worked well for low values of θ_d however beyond $\theta_d > 44°$ digits II and IV had a very small contact area with the cylinder.

4.2. Mechanical Improvements

Firstly to address the loss of contact area at large separation angles ($\theta_d > 44°$), a spherical substrate could have been tested as opposed to a cylindrical one. For a spherical substrate, as θ_d varies, each digit would still contact the sphere tangentially, which would have increased the repeatability of our tests.

From a design perspective, making the overall claw larger to facilitate the use of an extensor tendon, would have allowed us to alter the diameter of the joints to achieve more mechanical advantage, particularly in the 2nd and 3rd phalanges. This would have mitigated the issue of our motors being underpowered.

The vernier mechanism could be improved to use a dowel pin in place of an M2.5 bolt to lock the angle of the digits. Using M2.5 clearance holes meant there was too wide of a tolerance band on the holes of the vernier mechanism, meaning even with the bolt in place, the knuckles could still rotate roughly $\pm 1°$.

The primary limitation was insufficient actuation force from the N20 micromotors (298:1 gear ratio, 3.3kg·cm stall torque), which produced approximately 2.1 N per digit [see supplementary material 1]. This falls significantly short of the 64–86 N grip forces demonstrated as functional minimums for bird-inspired grippers [6]. Motors consistently reached stall torque when attempting to grip objects, yet failed to generate secure holds.

Contributing factors included substantial tendon friction losses during routing through the finger structure, with literature suggesting efficiency reductions of up to 37% in

similar systems [19]. Additionally, the mechanical advantage distribution was suboptimal, with the 1st phalange requiring disproportionate force to initiate closure of the 2nd and 3rd phalanges [20]. Future iterations should incorporate higher torque-density motors with integrated encoders, such as coreless brushed or brushless variants offering 2–3 times the power in comparable form factors. Furthermore, if this part were going into final production, we would recommend printing from a blended PTFE filament with a 0.08mm layer height since PTFE has a much lower coefficient of friction than PLA (0.05 vs 0.4).

4.3. Electrical and Software Improvements

4.3.1. Sensing Architecture

The haptic feedback system proved fundamentally inadequate, with the single palm-mounted FSR failing to provide actionable data for adaptive grasping. Force Sensing Resistors exhibit inherent limitations including drift errors of 15–30% over short timeframes, significant hysteresis, and high sensitivity to contact geometry and shear forces [21]. Figure 9 demonstrates these drift characteristics, showing measurement error increasing substantially over time, particularly for Interlink sensors which exhibited up to 30% drift after 270 seconds of static loading. Beyond these sensor characteristics, the single-point measurement provided no spatial information about grip distribution and could not detect object deformation or compliance.

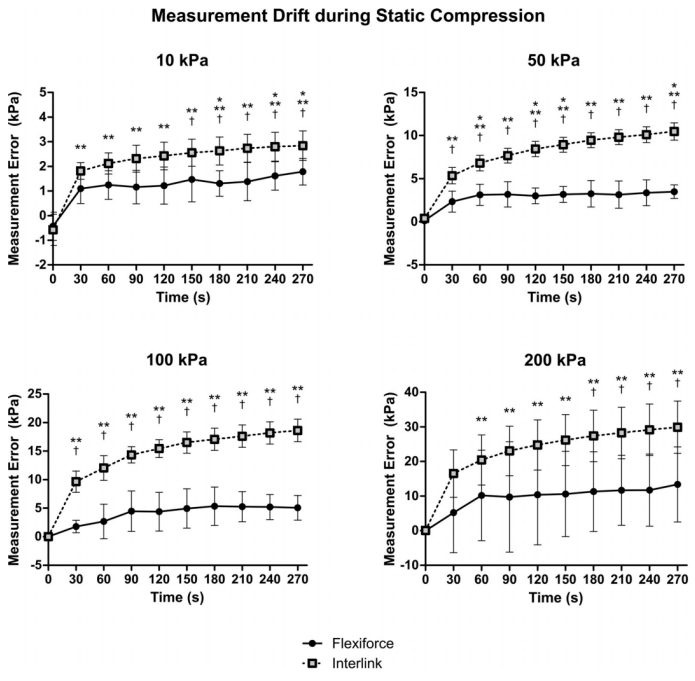


Figure 9: FSR measurement drift during static compression at various pressure levels, demonstrating significant error accumulation over time [21].

A robust solution requires a multi-modal sensing architecture combining complementary feedback mechanisms. Distributed pressure mapping arrays using piezoresistive materials such as Velostat or capacitive sensor matrices would provide spatial force distribution data, with contact area changes serving as indicators of object deformation [22]. High-resolution optical encoders would enable precise position feedback for each actuator, allowing detection of tendon slip and servo-like control. Motor current sensing via Hall-effect sensors could provide indirect torque estimation, though this approach requires careful signal filtering and observer-based estimation algorithms to mitigate noise and friction-induced errors [23]. Together, these modalities would enable real-time construction of force-displacement curves characterizing object compliance [24].

4.3.2. Vision-Based Intelligence

Incorporating computer vision with machine learning for pre-grasp assessment represents a significant enhancement opportunity. Vision-based object recognition using architectures such as YOLO for detection and ResNet for classification would enable identification of object type, material properties, and fragility indicators before physical contact [25]. This approach mirrors biological grasping strategies, as raptors employ targeted head movements to position their high-acuity central fovea for detailed visual assessment of prey prior to talon engagement [26].

The visual information would interface with an intelligent force database, allowing the system to adaptively select appropriate grip parameters based on object classification. This would enhance both safety (preventing damage to delicate or living materials) and versatility (optimizing forces for different material types and geometries). Implementation would require embedded processing capabilities for real-time inference.

4.3.3. Control Integration and Automation

The integration of feedforward control via vision with feedback control via tactile sensing offers substantial performance improvements, with literature demonstrating accuracy increases from 93% (vision alone) to 99.3% in multimodal fusion systems [25]. Additional control enhancements should address current operational limitations including the manual digit separation adjustment required between different object sizes.

Mechanically intelligent underactuated designs passively adapt grasp configuration based on object contact. Single-actuator systems utilizing compliant linkages, tendon routing, and joint springs can automatically transition between parallel fingertip grasps and enveloping power grasps depending on where contact occurs along the finger [10]. Figure 4 shows how a single motor drives both flexor and extensor tendons through routing points and joint springs, enabling autonomous grasp reconfiguration without active control of individual joints.

Furthermore, encoder-based position homing using photoelectric switches combined with encoder index pulses would eliminate the need for manual priming between grasp cycles, establishing consistent reference positions automatically [27]. These improvements collectively would transform the system from a proof-of-concept demonstration into a robust adaptive grasping platform capable of autonomous operation across diverse object types.

4.4. Applications

This claw design targets sectors requiring precise and delicate handling. The industrial automation industry relies heavily on robotic grippers which is displayed through the $1.79 billion market size (2024) [28], and is projected to grow at a compound annual growth rate (CAGR) of 18.8% between 2025-2033, reaching a market value of $6.3 billion [29]. Within this market growth, the food and beverages segment is projected to grow at a CAGR of 23.4% [28], driven by the need for flexible and delicate handling solutions. Similarly, biomedical and chemical industries would benefit from a design focused on minimising deformation and pressure during handling.

The claw's lightweight, adaptable design can also be implemented in other industries including archaeological exploration and retrieval along with agricultural harvesting and pest control.

Additionally, the exploration of optimal digit separation and adjustable digits holds significant value to biomimetic research and future developments of advanced grippers.

5. Conclusion

Inspired by secure avian grasping, we designed and constructed a prototype claw with a systematically variable digit-separation angle. The focus on studying the effects of angle separation on grasping mechanics explored a gap in academic research for which there could be interesting results and insights.

We have concluded that our experiments provided insufficient and unreliable data, and thus have failed to accept our hypothesis. This study revealed key challenges, particularly sensing limitations, low motor torque and tendon friction and offers suggested improvements on our design, such that further research could be carried out to make a more informed decision on the validity of the hypothesis. These developments would allow for more consistent and reliable data, advancing the understanding of adaptive grasping in biomimetic robotics.

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